COVID-19 Model: Scheme and Equations

1. Compartmental scheme

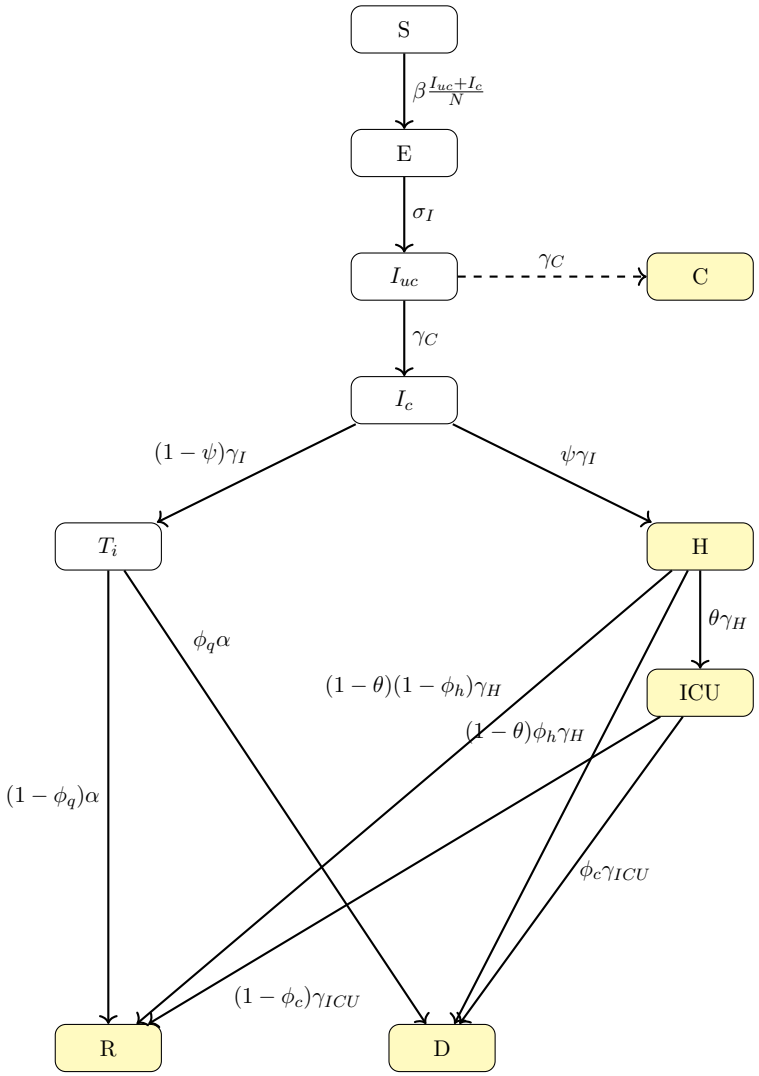


Figure 1: *
Yellow compartments are observed series: C (confirmed cases), H (ward), ICU, D (deaths). C is a non-depleting counter on the $I_{uc} \to I_c$ flow.

2. Model equations

$$
\begin{aligned}
\frac{dS}{dt} &= -\beta \frac{S\,(I_{uc} + I_c)}{N} \\
\frac{dE}{dt} &= \beta \frac{S\,(I_{uc} + I_c)}{N} - \sigma_I E \\
\frac{dI_{uc}}{dt} &= \sigma_I E - \gamma_C I_{uc} \\
\frac{dI_c}{dt} &= \gamma_C I_{uc} - \gamma_I I_c \\
\frac{dT_i}{dt} &= (1-\psi)\,\gamma_I I_c - \alpha\, T_i \\
\frac{dH}{dt} &= \psi\,\gamma_I I_c - \gamma_H H \\
\frac{d\,ICU}{dt} &= \gamma_H\,\theta H - \gamma_{ICU}\,ICU \\
\frac{dR}{dt} &= (1-\phi_q)\,\alpha T_i + (1-\phi_h)\,\gamma_H(1-\theta)\,H + (1-\phi_c)\,\gamma_{ICU}\,ICU \\
\frac{dD}{dt} &= \phi_q\,\alpha T_i + \phi_h\,\gamma_H(1-\theta)\,H + \phi_c\,\gamma_{ICU}\,ICU \\
\frac{dC}{dt} &= \gamma_C I_{uc}
\end{aligned}
$$

3. Testing-volume coupling

$$Ch(t) = \exp\!\left(-F_{\text{test}} \cdot \frac{NT(t)}{12\,632}\right)$$

where $NT(t)$ is the 7-day rolling sum of daily tests, and 12 632 is the mean of $NT(t)$ over 2020. $Ch(t) \in (0, 1]$ falls as testing volume grows, scaling down the hospitalisation probability below.

4. Hospitalisation probability $\psi(t)$

Model 1 (constant): $\psi(t) = \psi$

Model 2 (piecewise): $\psi(t) = \psi_k$ for t in segment k (one value per NPI-changepoint segment)

Model 3a (age-stratified, without testing-volume coupling):

$$\psi(t) = \sum_a P_a(t)\,\psi_a, \qquad a \in \{\text{0–49, 50–59, 60–69, 70+}\}$$

Model 3b (age-stratified, with testing-volume coupling):

$$\psi(t) = Ch(t) \cdot \sum_a P_a(t)\,\psi_a$$

In both age-stratified models,

$$P_a(t) = \frac{c_a(t)}{\sum_b c_b(t)}, \qquad \psi_{\text{50-59}} = 2\,\psi_{\text{0-49}}, \quad \psi_{\text{60-69}} = 4\,\psi_{\text{0-49}}, \quad \psi_{\text{70+}} = 8\,\psi_{\text{0-49}}$$

where $c_a(t)$ is the number of age-labelled confirmed cases in group a on day t, and $\psi_{\text{0-49}}$ is the single estimated base rate.